

CS 101 - Algorithms & Programming I

Spring 2026 - Lab 4

Due: Week of March 2

Remember the honor code for your programming assignments.

For all labs, your solutions must conform to the CS101 style guidelines!

All data and results should be stored in variables (or constants where appropriate) with meaningful names.

The objective of this lab is to learn how to use **for** and **while** loops to implement automated repetition. Remember that analyzing your problems and designing them on a piece of paper *before* starting implementation/coding is always a best practice.

0. Setup Workspace

Start VSC and open the previously created workspace. Create a new folder named `lab4`.

In this lab, you are to have three Java classes/files for the main questions as described below. The fourth and the fifth file containing the revisions should go under this folder as well. We expect you to submit a total of 5 files, including the revisions, **without compressing** them. Do *not* upload other/previous lab solutions in your submission. The user inputs in the sample runs are shown in [blue](#).

1. Hourglass Shape

Create a new file under the `lab4` folder named `Lab04_Q1.java` with a class with the same name.

In this program, you are expected to print an **"Hourglass"** shape with stars. To decide the size of the hourglass, you will get an **odd integer** input from the user (representing the width of the widest part).

- You are supposed to check the input and proceed only if it is a **positive odd number**.
- You should ask for input repeatedly using a **while** loop until a valid number is given.
- Your task is to print the hourglass pattern using nested **for** loops. The shape consists of an inverted triangle on top and a regular triangle on the bottom.

Sample run

```
Enter an odd number for the hourglass pattern: -5
Invalid input, try again.
Enter an odd number for the hourglass pattern: 8
Invalid input, try again.
Enter an odd number for the hourglass pattern: 7
*****
 *
 ***
  *
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```

2. Palindrome Check

Create a new file under the `lab4` folder named `Lab04_Q2.java` with a class with the same name.

In this program, you are expected to print if the input string is a **palindrome** or not. Palindrome is a word or phrase that reads the same backwards as forwards.

- You are supposed to receive inputs as strings and utilize string functions on them.
- You should program this as case-sensitive, e.g. "a" and "A" are considered as different characters.
- You should ask for input repeatedly using a **while** loop.
- The program ends when the user inputs 0.

Sample run

```
Enter a string: 121
It is a palindrome.
Enter a string: derya
It is not a palindrome.
Enter a string: kabak
It is a palindrome.
Enter a string: Kabak
It is not a palindrome.
Enter a string: 0
Program terminated.
```

3. The Savings Race

Create a new file under the `lab4` folder named `Lab04_Q3.java`, containing a class with the same name.

Your task is to implement a financial simulation called "**The Savings Race**". Three students (Student A, Student B, and Student C) are competing to see who can save enough money to buy a new laptop first.

Simulation Overview

- **Target:** The goal is to reach a savings balance of **\$1,000.00**.
- **Players:** There are 3 students. Each starts with a different **Initial Balance** and saves a different **Monthly Amount**.
- **The Loop:** The simulation runs month-by-month. In every "round" (month), you must calculate the new balance for all three students.
- **Winning:** The simulation stops as soon as **one or more** students reach or exceed the target amount.

Game Flow

1. **Setup:**
 - Define the goal: **\$1000.00**.
 - Initialize the data for the 3 students (initialize in the code as variables, not as user inputs):
 - **Student A:** Starts with \$100.00, saves \$50.00/month.
 - **Student B:** Starts with \$300.00, saves \$30.00/month.
 - **Student C:** Starts with \$0.00, saves \$75.00/month.
2. **Monthly Loop:**
 - Use a **while** loop to simulate time.
 - Inside the loop, increase each student's balance by their monthly savings amount.
 - Print the status of the current month.

3. End Condition:

- Check if any student has reached **\$1000.00**.
- If yes, break the loop.
- Print the total months taken and the final balances.
- Print the winner. If multiple students pass the threshold in the same month, print all of them.

Sample run

```
Welcome to the Savings Race!
Target: $1000.00

Month 1:
Student A: 150.00 | Student B: 330.00 | Student C: 75.00
Month 2:
Student A: 200.00 | Student B: 360.00 | Student C: 150.00
Month 3:
Student A: 250.00 | Student B: 390.00 | Student C: 225.00
Month 4:
Student A: 300.00 | Student B: 420.00 | Student C: 300.00
Month 5:
Student A: 350.00 | Student B: 450.00 | Student C: 375.00
Month 6:
Student A: 400.00 | Student B: 480.00 | Student C: 450.00
Month 7:
Student A: 450.00 | Student B: 510.00 | Student C: 525.00
Month 8:
Student A: 500.00 | Student B: 540.00 | Student C: 600.00
Month 9:
Student A: 550.00 | Student B: 570.00 | Student C: 675.00
Month 10:
Student A: 600.00 | Student B: 600.00 | Student C: 750.00
Month 11:
Student A: 650.00 | Student B: 630.00 | Student C: 825.00
Month 12:
Student A: 700.00 | Student B: 660.00 | Student C: 900.00
Month 13:
Student A: 750.00 | Student B: 690.00 | Student C: 975.00
Month 14:
Student A: 800.00 | Student B: 720.00 | Student C: 1050.00

Race finished!
Total months: 14
Final Balances:
Student A: $800.00
Student B: $720.00
Student C: $1050.00

Winner(s):
Student C
```